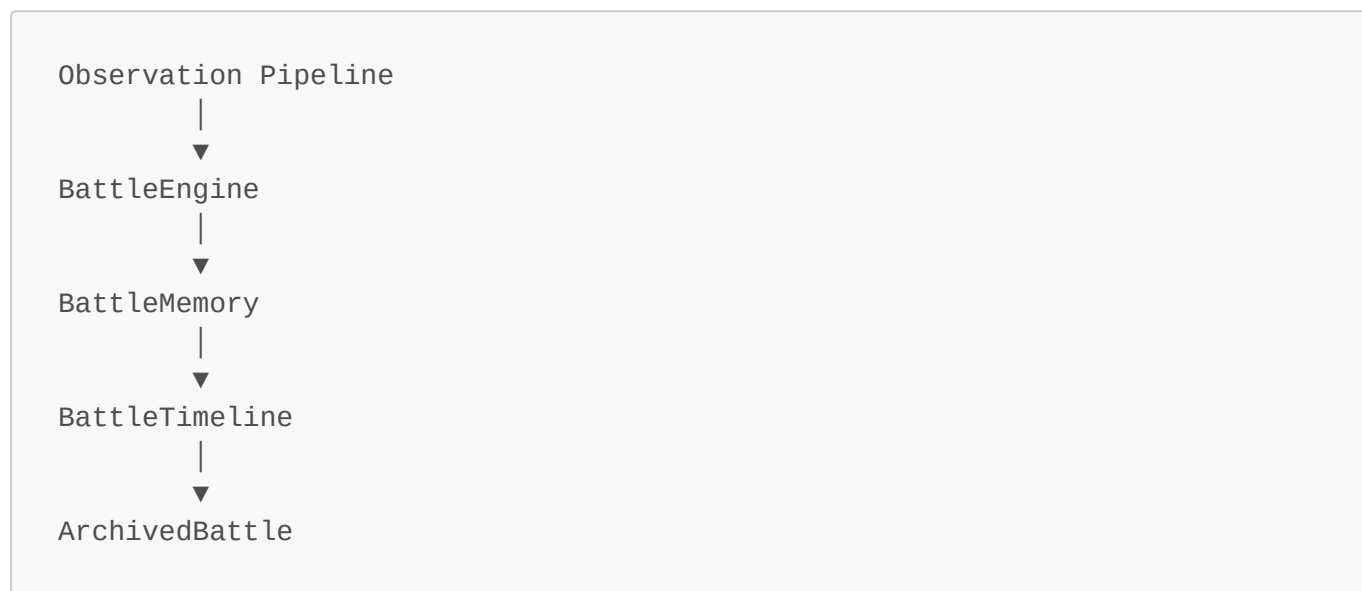


BattleEngine

The **BattleEngine** coordinates the entire data lifecycle within Overdex, ensuring that raw observations are correctly transformed into a structured record of combat.

Data Lifecycle



Responsibilities

- **Coordination:** Managing the battle lifecycle from **BATTLE_STARTED** to **BATTLE_ENDED**.
- **Normalization:** Translating varied inputs (OCR, Audio, Manual) into stable internal state changes.
- **Auditing:** Ensuring that every meaningful transition is recorded as a fact in the chronological timeline.
- **Persistence Packaging:** Facilitating the transition from live, mutable memory to a permanent, immutable **ArchivedBattle** snapshot.

Intelligence Layers

The engine provides data to analytical layers that interpret the battle without modifying the record:

1. **Matchup Engine:** Evaluates the tactical relationship between the active Pokémon.
2. **Decision Engine:** Translates matchup data into high-level player recommendations.

Design Philosophy

Invisible decision support.

The engine quietly observes and remembers so the player can focus on gameplay. It surfaces only the information that reduces cognitive load, answering the core question: **"What decision does the player need to make right now?"**